

Enforcing Thermodynamic Invariants in Planetary Biogeochemical Simulations: The Thermodynamic State Vector Stock Conservation Asserter

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Sprint 054 – Repository: <https://github.com/pascalranoroarijaona/WebOfLife>

Abstract

Planetary-scale ecological and biogeochemical simulations frequently encounter numerical drift, violating fundamental physical conservation laws over extended temporal integration horizons. This paper details the architectural implementation and rigorous mathematical formulation developed in Sprint 054: the **Thermodynamic State Vector Stock Conservation Asserter** (`src/thermodynamics/state_validator.ts`). By continuously evaluating discrete state transitions against integrated boundary fluxes across carbon, nitrogen, phosphorus, water, and energy pools, our system enforces strict mathematical adherence to the First Law of Thermodynamics (mass-energy conservation) and the Second Law (entropy generation and solar-only external forcing constraints). We present the formal mathematical framework, the executable TypeScript monad validation routine, and discuss its implications for maintaining stable, non-spontaneous planetary ecosystems within the Web of Life simulation engine.

1 Introduction & Systems Ecology Context

In complex systems ecology and Earth system modeling, simulating biogeochemical cycles requires tracking planetary state vectors across millions of discrete time steps. Without rigorous mathematical constraints, numerical integration errors accumulate, leading to unphysical phenomena such as phantom mass creation or spontaneous energy generation.

Within the **Web of Life** architecture, state vectors represent continuous ecological stocks, while functional reactive monads manage state transitions. Sprint 054 introduces an automated guardian—the **StateValidator**—to intercept state updates and assert physical validity before state commitment.

2 Mathematical Formulation of Conservation

2.1 First Law Conservation Equation

For any conserved elemental stock or energy pool i (where $i \in \{\text{Carbon, Nitrogen, Phosphorus, Water, Energy}\}$), the true physical invariant requires that the change in stock mass/energy within the system boundaries exactly equals the net flux across those boundaries over time interval $\Delta t = t_1 - t_0$:

$$\Delta S_i = S_i(t_1) - S_i(t_0) = \int_{t_0}^{t_1} \left(\sum \Phi_{i,\text{in}}(t) - \sum \Phi_{i,\text{out}}(t) \right) dt \quad (1)$$

For discrete numerical implementation with constant or piecewise linear fluxes over interval Δt :

$$\Delta S_{i,\text{actual}} = S_i(t_1) - S_i(t_0) \quad (2)$$

$$\Delta S_{i,\text{expected}} = \left(\sum \Phi_{i,\text{in}} - \sum \Phi_{i,\text{out}} \right) \cdot \Delta t \quad (3)$$

The validation residual ε_i is tested against a predefined numerical tolerance τ_i :

$$|\Delta S_{i,\text{actual}} - \Delta S_{i,\text{expected}}| \leq \tau_i \quad (4)$$

2.2 Second Law & Boundary Invariants

1. **No Spontaneous Generation:** If $\sum \Phi_{i,\text{in}} = 0$ and $\sum \Phi_{i,\text{out}} = 0$, then $|\Delta S_{i,\text{actual}}| \leq \tau_i$.
2. **Solar-Only External Energy Forcing:** External energy influx $\Phi_{E,\text{in}}$ must originate exclusively from designated solar irradiance boundary vectors (Q_{solar}). Any unassigned internal energy generation violates entropy bounds and triggers a `ThermodynamicViolationException`.

3 Software Architecture & Implementation

The assserter is housed in `src/thermodynamics/state_validator.ts`. It provides rule registration and execution routines integrated into the main thermodynamic monad loop:

```
import { StateVector } from './state_vector';
import { ThermodynamicFlux, ValidationResult, ThermodynamicViolationException } from './types';

export class StateValidator {
  private rules: Map<string, number> = new Map();

  constructor(defaultTolerance: number = 1e-5) {
    this.rules.set('carbon', defaultTolerance);
    this.rules.set('nitrogen', defaultTolerance);
    this.rules.set('phosphorus', defaultTolerance);
    this.rules.set('water', defaultTolerance);
    this.rules.set('energy', defaultTolerance);
  }

  public validateStockConservation(
    previousState: StateVector,
    currentState: StateVector,
    boundaryFluxes: ThermodynamicFlux[],
    deltaTime: number
  ): ValidationResult {
    // 1. Calculate actual stock delta: Delta Stock = Stock(t_1) - Stock(t_0)
    // 2. Accumulate boundary fluxes over deltaTime: NetFlux = Influx - Outflux
    // 3. Assert |Delta Stock - NetFlux * dt| <= tolerance
    // 4. Enforce First & Second Law invariants
  }
}
```

4 Conclusion

Sprint 054 establishes a robust thermodynamic foundation for the Web of Life simulator, ensuring that all biogeochemical dynamics respect physical reality. Future sprints will extend this framework to multi-node spatial diffusion and localized exergy dissipation profiling.

Official Repository: <https://github.com/pascalranoroarijaona/WebOfLife>